Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE

Logic Design and Microprocessor

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Assignment 2:

TITLE:- K-map-e.g. Code converters excess-3 to BCD and vice versa using logical gates

Description:- K-Map Based Code Converter (Excess-3 to BCD and BCD to Excess-3 Using Logic Gates)
1. What is K-Map?

A Karnaugh Map (K-Map) is a graphical method used to simplify Boolean expressions. It helps reduce the number of logic gates required in a digital circuit by minimizing Boolean functions. K-Maps are widely used in digital electronics for designing combinational logic circuits such as code converters, adders, multiplexers, and decoders.

Advantages of K-Map:

Simplifies Boolean expressions.
Reduces the number of logic gates.
Minimizes hardware cost.
Improves circuit efficiency.
Easy to implement for up to 4–6 variables.
2. What is BCD (Binary Coded Decimal)?

BCD (Binary Coded Decimal) is a coding system in which each decimal digit (0–9) is represented by its corresponding 4-bit binary number.

BCD Code Table

Decimal	BCD
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Only these ten combinations are valid in BCD.

3. What is Excess-3 (XS-3) Code?

Excess-3 (XS-3) is a 4-bit self-complementing code obtained by adding 3 (0011) to each decimal digit before converting it into binary.

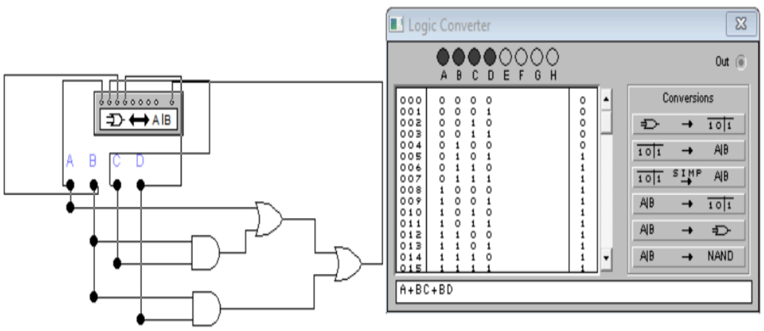
Excess-3 Code Table

Decimal	BCD	Excess-3
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

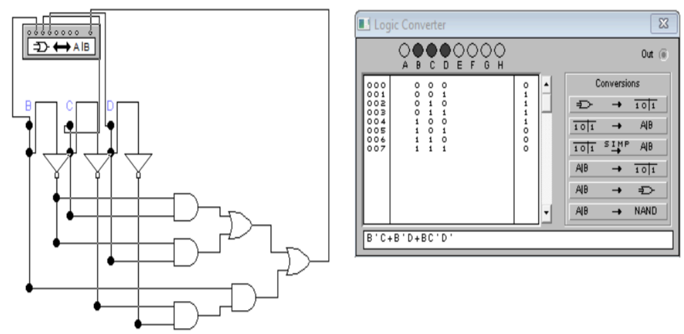
- **Circuit Diagram**

From BCD to excess-3

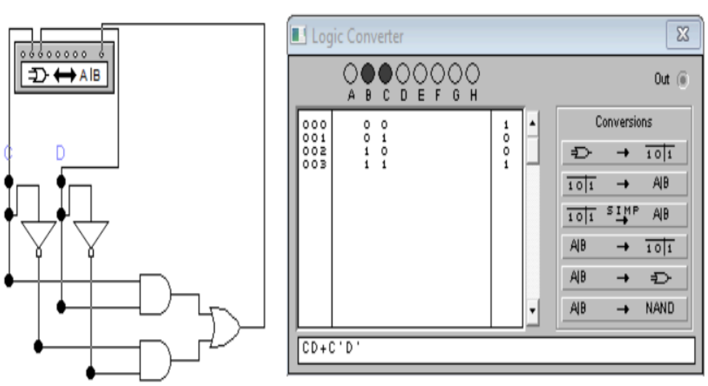
W=A+BC+BD



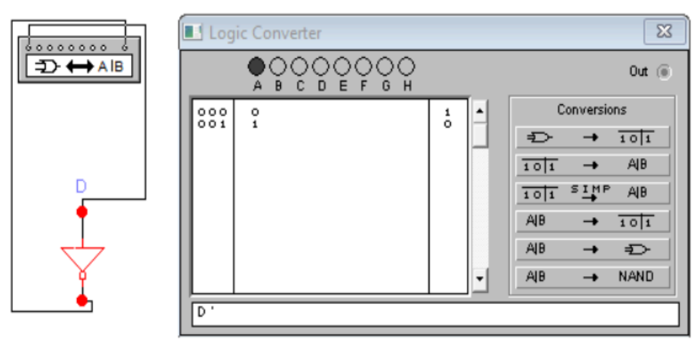
X=B’C+B’D+BC’D’



Y=CD+C’D’

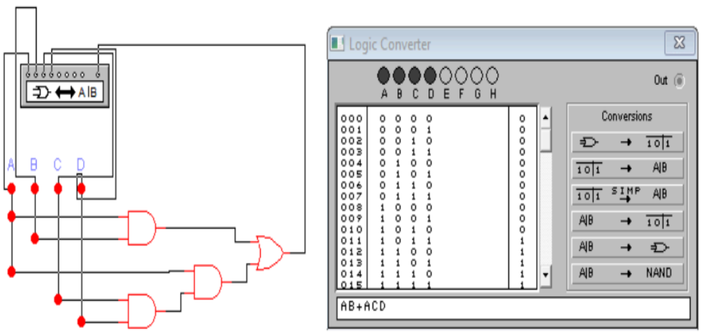


Z=D’

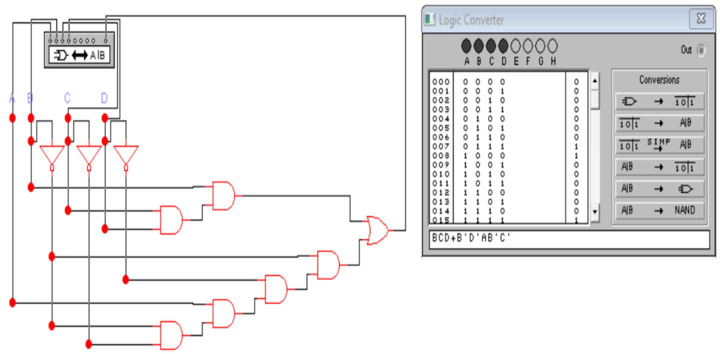


From excess-3 to BCD

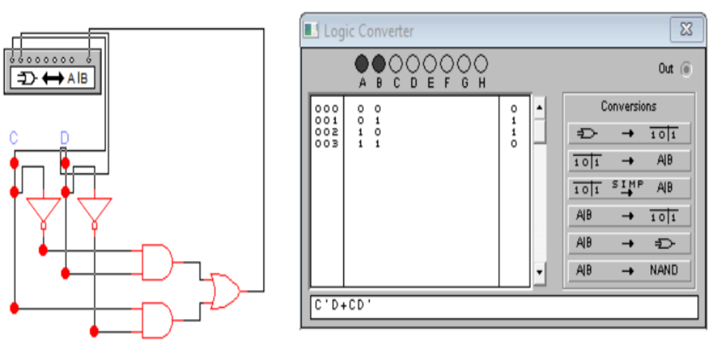
W=AB+ACD



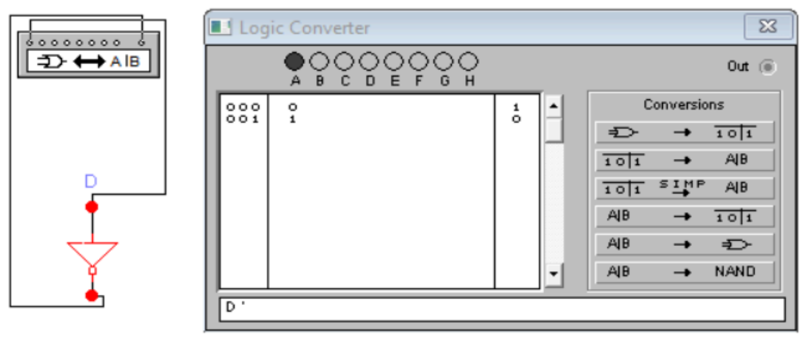
X=BCD+B’D’+AB’C’



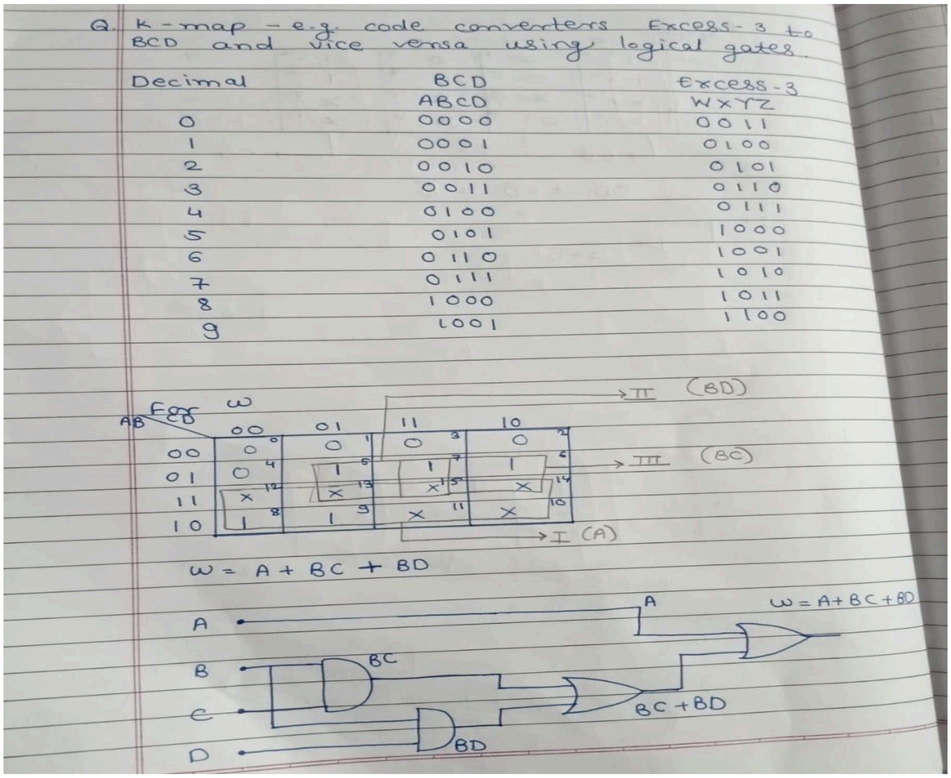
Y=C’D+CD’



Z=D’



- **Screenshots/Output:**



Q. K-map – e.g. code converters Excess-3 to BCD and vice versa using logical gates.

Decimal	BCD ABCD	Excess-3 WXYZ
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

For W

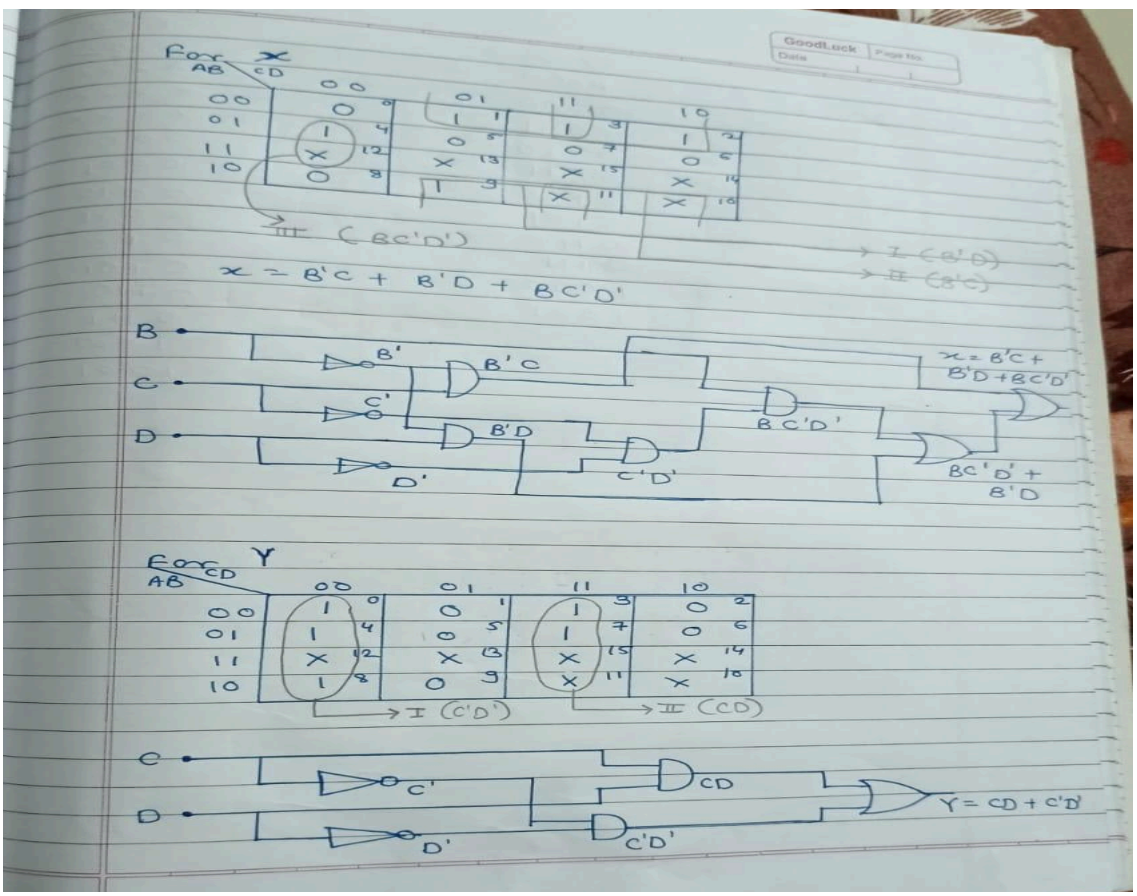
AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

II (BD)

III (BC)

I (A)

W = A + BC + BD



For X

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

III (BC'D')

I (B'D)

II (B'C)

x = B'C + B'D + BC'D'

For Y

AB \ CD	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	X	X	X	X
10	1	0	X	X

I (C'D')

II (CD)

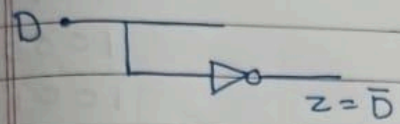
For Z

AB \ CD	00	01	11	10
00	1 (0)	0 (1)	0 (3)	1 (2)
01	1 (4)	0 (5)	0 (7)	1 (6)
11	X (12)	X (13)	X (15)	X (14)
10	1 (8)	0 (9)	X (11)	X (10)

→ $\bar{C}\bar{D}$

→ $C\bar{D}$

$$\bar{C}\bar{D} + C\bar{D} = \bar{D}$$



Decimal	excess-3 ABCD	BCD WXYZ
0	0000	XXXX
1	0001	XXXX
2	0010	XXXX
3	0011	0000
4	0100	0001
5	0101	0010
6	0110	0011
7	0111	0100
8	1000	0101
9	1001	0110
10		0111
11		1000
12		1001

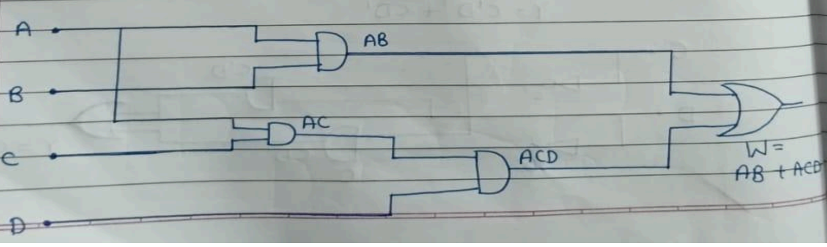
For W

AB \ CD	00	01	11	10
00	X (0)	X (1)	0 (3)	X (2)
01	0 (4)	0 (5)	0 (7)	0 (6)
11	1 (12)	X (13)	X (15)	X (14)
10	0 (8)	0 (9)	1 (11)	0 (10)

→ (AB) I

→ II (ACD)

W = AB + ACD

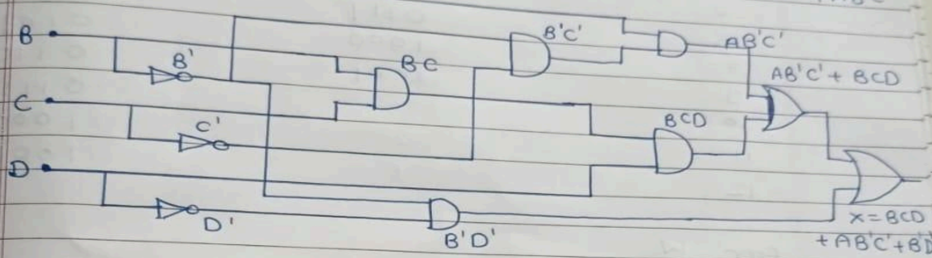


For X

AB \ CD	00	01	11	10
00	X (0)	X (1)	0 (3)	X (2)
01	0 (4)	0 (5)	1 (7)	0 (6)
11	0 (12)	X (13)	X (15)	X (14)
10	1 (8)	1 (9)	0 (11)	1 (10)

→ III (BCD)
→ II (AB'C')
→ I (B'D')

X = BCD + B'D' + AB'C'

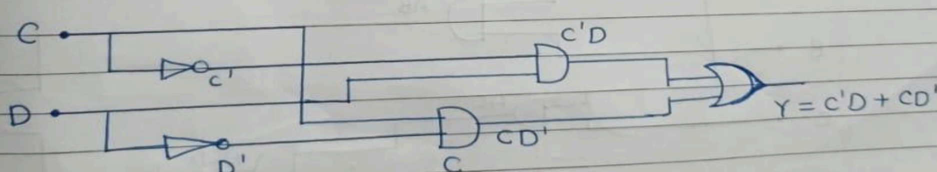


For Y

AB \ CD	00	01	11	10
00	X (0)	X (1)	0 (3)	X (2)
01	0 (4)	ϕ (5)	0 (7)	ϕ (6)
11	0 (12)	X (13)	X (15)	X (14)
10	0 (8)	1 (9)	0 (11)	1 (10)

→ II (C'D)
→ I (CD')

Y = C'D + CD'

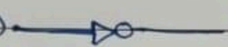


for Z

AB \ CD	00	01	11	10
00	X [0]	X [1]	0 [3]	X [2]
01	1 [4]	0 [5]	0 [7]	1 [6]
11	1 [12]	X [13]	X [15]	X [14]
10	1 [8]	0 [9]	0 [11]	1 [10]

I $(C'D')$ II (CD')

$$Z = C'D' + CD'$$
$$= D'(C'+C)$$
$$= D'$$

D  $Z = D'$